Enterprise Supply Chain Intelligence

1. Project Overview

Enterprise Supply Chain Intelligence is an industry-level Power BI analytical solution for a logistics and fleet management business. The dashboard is designed to help management and operations teams understand overall logistics performance, revenue and shipment performance, delivery service levels, fleet utilization, driver performance, fuel consumption, maintenance, safety, customer performance, and route performance.

The analytical workflow is: EDA → Data Cleaning → Power Query → Data Modelling → DAX → Visualization → Business Insights.

The source specification explicitly requires a dimensional model and emphasizes correct data, correct grain, correct model, correct DAX, and correct business interpretation.

2. Business Problem

Logistics operations generate information across loads/trips, delivery events, fuel purchases, maintenance events, safety incidents, drivers, trucks, customers, routes, and facilities. The business needs a consistent analytical layer that can combine these processes without incorrectly merging different grains. The dashboard therefore focuses on operational visibility, performance measurement, trend analysis, exception identification, and data-supported business insights.

3. Project Objectives

Objective	Description
Overall performance	Monitor the overall health of logistics operations.
Revenue & shipments	Understand revenue, loads, trips, miles, and shipment activity.
Delivery service	Measure on-time delivery and detention performance.
Fleet efficiency	Analyze truck utilization, miles, MPG, and revenue.
Driver performance	Compare driver productivity, efficiency, and safety.
Fuel & cost	Analyze fuel consumption, fuel price, and fuel cost.
Maintenance reliability.	Understand maintenance expenditure, downtime, and asset
Safety	Monitor incidents, preventability, damage, and claim exposure.
Customer & route	Identify customer value and route performance patterns.

4. Data Architecture

The required architecture is a Star Schema / Dimensional Model. Dimension tables provide descriptive context, while fact tables represent separate business processes.

4.1 Dimension Tables

Table	Purpose
DimDate	Common date dimension for time analysis and time intelligence.
DimCustomer	Customer attributes and customer segmentation.
DimDriver	Driver attributes and employment/operational context.
DimTruck	Truck/unit attributes and fleet context.
DimTrailer	Trailer attributes.
DimRoute	Route definitions and route context.
DimFacility	Facility/location attributes.

4.2 Fact Tables

Table	Business Process	Required Grain
FactLoadTrip	Loads / trips	1 row = 1 load/trip
FactDeliveryEvent	Pickup / delivery events	1 row = 1 pickup/delivery event
FactFuelPurchase	Fuel purchases	1 row = 1 fuel purchase
FactMaintenance	Maintenance activity	1 row = 1 maintenance event
FactSafetyIncident	Safety incidents	1 row = 1 safety incident
FactDriverMonthly	Driver monthly aggregate	1 row = 1 driver/month
FactTruckMonthly	Truck monthly aggregate	1 row = 1 truck/month

Important modeling rule: Do not create a single giant table by merging all fact tables. Maintain the correct grain of each business process.

5. Data Dictionary

The project specification requires a data dictionary containing the column name, table, data type,description, and business meaning for important columns. The supplied project specification defines the tables and business processes, but it does not contain the actual source Excel column inventory.Therefore, the following is a business-level data dictionary; exact physical column names and data types should be filled from the final dataset before submission.

5.1 FactLoadTrip

Business Field	Table	Expected Type	Description / Business Meaning
Load / Trip ID	FactLoadTrip	Text / Key	Unique identifier for the load or trip.
Date Key	FactLoadTrip	Date / Key	Links the load/trip to DimDate.
Customer Key	FactLoadTrip	Key	Links the load/trip to DimCustomer.
Driver Key	FactLoadTrip	Key	Links the load/trip to DimDriver.
Truck Key	FactLoadTrip	Key	Links the load/trip to DimTruck.
Route Key	FactLoadTrip	Key	Links the load/trip to DimRoute.
Load Type	FactLoadTrip	Categorical	Classification of the load.
Booking Type	FactLoadTrip	Categorical	How the load was booked.
Revenue	FactLoadTrip	Numeric	Revenue associated with the load/trip.
Miles	FactLoadTrip	Numeric	Miles associated with the load/trip.

5.2 FactDeliveryEvent

Business Field	Table	Expected Type	Description / Business Meaning
Event ID identifier.	FactDeliveryEvent	Text / Key	Unique delivery/pickup event
Date Key	FactDeliveryEvent	Date / Key	Links event activity to DimDate.
Customer Key	FactDeliveryEvent	Key	Customer associated with the event.
Route Key	FactDeliveryEvent	Key	Route associated with the event.
Facility Key	FactDeliveryEvent	Key	Facility associated with the event.
Event Type classification.	FactDeliveryEvent	Categorical	Pickup or delivery event
Scheduled	FactDeliveryEvent	DateTime	Scheduled event timing.
Date/Time			
Actual	FactDeliveryEvent	DateTime	Actual event timing.
Date/Time			
Detention	FactDeliveryEvent	Numeric	Detention duration in minutes.
Minutes			
On-Time Flag	FactDeliveryEvent	Boolean / Flag	Indicates whether an applicable
			event was on time.

5.3 FactFuelPurchase

Business Field	Table	Expected Type	Description / Business Meaning
Fuel Purchase ID	FactFuelPurchase	Text / Key	Unique fuel purchase identifier.
Date Key	FactFuelPurchase	Date / Key	Fuel purchase date reference.
Truck Key	FactFuelPurchase	Key	Truck associated with the purchase.
Driver Key	FactFuelPurchase	Key	Driver associated with
			the purchase, where applicable.
State	FactFuelPurchase	Categorical	State/location associated
			with purchase.Business
Fuel Type	FactFuelPurchase	Categorical	Type of fuel purchased.
Fuel Gallons	FactFuelPurchase	Numeric	Quantity of fuel purchased.
Fuel Cost	FactFuelPurchase	Numeric	Total purchase cost.
Price per Gallon	FactFuelPurchase	Numeric	Average/transaction fuel price per gallon.

5.4 FactMaintenance

Business Field	Table	Expected Type	Description / Business Meaning
Maintenance Event ID identifier.	FactMaintenance	Text / Key	Unique maintenance event
Date Key reference.	FactMaintenance	Date / Key	Maintenance event date
Truck Key	FactMaintenance	Key	Truck receiving maintenance.
Maintenance Type	FactMaintenance	Categorical	Type/category of maintenance.
Maintenance Cost maintenance.	FactMaintenance	Numeric	Cost associated with
Downtime Hours	FactMaintenance	Numeric	Asset downtime associated with
			the event.
Facility Location	FactMaintenance	Categorical	Maintenance facility/location.

5.5 FactSafetyIncident

Business Field	Table	Expected Type	Description / Business Meaning
Incident ID	FactSafetyIncident	Text / Key	Unique safety incident identifier.
Date Key	FactSafetyIncident	Date / Key	Incident date reference.
Driver Key	FactSafetyIncident	Key	Driver associated with incident.
Truck Key	FactSafetyIncident	Key	Truck associated with incident.
Incident Type	FactSafetyIncident	Categorical	Classification of safety incident.
Preventable Flag	FactSafetyIncident	Boolean / Flag	Indicates preventable vs
			non-preventable.
Injury Flag occurred.	FactSafetyIncident	Boolean / Flag	Indicates whether an injury
Vehicle Damage Cost	FactSafetyIncident	Numeric	Vehicle damage financial impact.
Cargo Damage Cost	FactSafetyIncident	Numeric	Cargo damage financial impact.
Claim Amount	FactSafetyIncident	Numeric	Associated claim amount.

5.6 Aggregate Fact Tables

Table	**Business MeaningExpected**	**Grain**
FactDriverMonthly	Monthly driver-level performance such as trips,	1 driver / month
	miles, revenue,MPG, fuel gallons, and on-time	
	delivery.	
FactTruckMonthly	Monthly truck-level performance and fleet metrics.	1 truck / month

5.7 Dimension Data Dictionary Summary

Dimension	**Important Business Attributes**
DimDate	Date, Year, Month, YearMonth and other calendar attributes required for
	time analysis.
DimCustomer	Customer identity, customer type, and active/customer classification.
DimDriver	Driver identity, employment status, home terminal, CDL class.
DimTruck	Truck/unit number, truck status, fuel type, home terminal.
DimTrailer	Trailer identity and trailer attributes.
DimRoute	Route identity and origin/destination context.
DimFacility	Facility identity, location and state context.

6. KPI Dictionary

Every KPI should have a clear definition, formula, source table, and business purpose. The following definitions are based on the supplied project specification.

KPI	Definition / Formula	Primary Source	Business Purpose
Total Revenue	Sum of revenue.	FactLoadTrip	Measure business revenue.
Total Loads	Count of load/trip records at the defined grain	FactLoadTrip	Measure shipment volume.
Total Trips	Count of trips at the defined Business grain.	FactLoadTrip	Measure trip activity.
Total Miles	Sum of miles.	FactLoadTrip	Measure operational distance.
Revenue per Mile	Total Revenue ÷ Total Miles.	FactLoadTrip	Assess revenue generated per mile.
Revenue per Load	Total Revenue ÷ Total Loads.	FactLoadTrip	Assess average revenue Per load.
Total Fuel Gallons	Sum of fuel gallons	FactFuelPurchase	Measure fuel consumption.
Total Fuel Cost	Sum of fuel cost.	FactFuelPurchase	Measure fuel expenditure.
Fuel Cost per Mile	Total Fuel Cost ÷ Total Miles.	Fuel + Load/Trip	Assess fuel cost efficiency.
Average MPG	Average / business-defined MPG metric.	Fleet/Truck data	Assess fuel efficiency.
Average Fuel Price	Fuel cost ÷ fuel gallons, or defined average price metric.	FactFuelPurchase	Monitor fuel price.
Total Maintenance cost	Sum of maintenance cost.	FactMaintenance	Measure maintenance expenditure
Total Detection Minute	Sum of detection Minute	Fact Delivery Event	Measure Operational Waiting Time

7. DAX Measure Approach

The project should use reusable DAX measures rather than repeatedly aggregating raw columns inside visuals.

Measure	Example DAX
Total Revenue	Total Revenue = SUM(FactLoadTrip[Revenue])
Total Loads	Total Loads = COUNTROWS(FactLoadTrip)
Total Miles	Total Miles = SUM(FactLoadTrip[Miles])
Revenue per Mile	Revenue per Mile = DIVIDE([Total Revenue], [Total Miles])
Revenue per Load	Revenue per Load = DIVIDE([Total Revenue], [Total Loads])

8. Time Intelligence

Required time-intelligence measures include Revenue MTD, Revenue YTD, Previous Year Revenue, Revenue YoY %, Loads YTD, and Miles YTD. These measures should only be created after DimDate is correctly configured.

9. Dashboard Documentation

Page	Purpose	Main Analytical Areas
1. Executive Overview	Senior-management overview.	Revenue, loads, trips, miles,
		fuel cost, revenue trends,
		revenue segments.
2. Shipment & Delivery	Operational shipment and delivery	On-time %, detention,
	performance.	transit duration, routes
		customers, facilities.
3. Fleet Performance	Vehicle utilization and efficiency.	Trucks, miles, MPG, revenue,
		truck status, monthly
		fleet performance.
4. Driver Performance	Driver productivity and operational	Drivers, trips, miles, revenue,
	Performance	MPG, safety incidents.
5. Fuel & Cost	Fuel consumption and operating cost.	Gallons, fuel cost, price/gallon
		, MPG, truck/state cost.
6. Maintenance	Maintenance expenditure and asset Health.	Maintenance cost, events, downtime,
		truck/type analysis.
7. Safety & Risk	Operational safety and financial	Incidents, preventability,
	exposure.	Injuries damage, claims.
8. Customer & . Route	Customer value and route. Performance	Customers, routes, revenue, loads, miles, on-time .

10. Advanced Power BI Features

Feature	Implementation / Purpose
Drillthrough	Truck Detail page from Fleet Performance; show revenue, miles, MPG, fuel
	cost, maintenance,downtime, safety incidents and monthly trend.
Report Tooltip truck.	Custom Truck Tooltip displaying key truck metrics when hovering over a
Dynamic Titles	Examples include Revenue Performance - Selected Year and Fleet
	Performance - Selected Truck.
Conditional Formatting	Highlight low on-time %, high maintenance cost, low MPG, or high incident count where useful.
Navigation	Use buttons to navigate between the eight analytical pages.

11. Business Insight Framework

The project specification requires at least 3–5 meaningful insights per major page. Insights must be supported by the data. Weak Insight Improved Insight Pattern Revenue is 10 million. Revenue increased during a period while revenue per mile declined, indicating that additional revenue was accompanied by disproportionately higher mileage. Truck X has high revenue. Truck X generates high revenue but also has significantly higher maintenance cost and downtime than comparable trucks, making it a candidate for asset-health review.

12. Business Recommendations Framework

Recommendations should be derived only after validating the underlying KPI and the supporting dimensions. Examples include investigating low-performing routes, reviewing assets with high maintenance and downtime, examining fuel-efficiency outliers, and reviewing recurring delivery detention patterns.

13. Data Quality & Limitations

The final implementation should document missing values, duplicate records, unusual values, data-type issues, inconsistent categories, relationship issues, and any business-definition limitations discovered during EDA and Power Query .Do not remove rows merely because values look unusual, fill every blank with zero, create unnecessary calculated columns, or make claims that cannot be supported by the dataset.

14. Final Deliverables

Deliverable	Expected Output
Power BI File	.pbix report containing the completed analytical dashboard.
Data Model	Screenshot/documentation showing fact tables, dimension
	tables, relationships, cardinality and filter direction.
Data Dictionary	Important columns with column name, table, data type, description
	and business meaning.
KPI Dictionary	KPI name, definition, formula, source table and business purpose.
Dashboard	8 analytical pages.
Insights	At least 3–5 meaningful insights per major analytical page.
Presentation	Business problem, dataset, EDA, cleaning, model, DAX, dashboard,
	insights, recommendations and limitations.

15. Interview / Presentation Preparation

The project should be explainable end-to-end: why the business problem matters, how the data was prepared, why the star schema was selected, how grain was maintained, how KPIs were defined, why particular visuals were selected, how findings were validated, and what limitations remain.

16. Important Modeling Rules

Avoid	Reason
Merging all fact tables	Can create incorrect grain and duplicate measures.
Unjustified many-to-many relationships	Can create ambiguous or incorrect filtering.
Bidirectional relationships everywhere	Can create unnecessary ambiguity and performance issues.
Many unnecessary calculated columns	Adds complexity when measures or transformations are more appropriate.
Separate date table for every fact approach.	The specification requires a common DimDate
Raw columns instead of reusable Measures	Reduces consistency and reusability.
Charts without a business question	Every visual should answer a business question

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17. Conclusion

Enterprise Supply Chain Intelligence provides a structured Power BI framework for analyzing logistics and fleet operations across revenue, shipments, delivery performance, fleet, drivers, fuel, maintenance, safety, customers, and routes. The quality of the final solution depends on maintaining the correct data grain, building the correct dimensional model, developing reusable DAX measures, selecting meaningful visuals, and generating business insights that are supported by the underlying data.

18. Source Scope Note

This document was prepared from the supplied 24-page project specification. The specification defines the required business scope, tables, grains, KPIs, dashboard pages, advanced features, deliverables, and evaluation expectations. It does not provide the actual source dataset's complete physical column list, row count, or measured results. Therefore, exact dataset-specific values and physical column names should be inserted after the final Excel dataset is available.